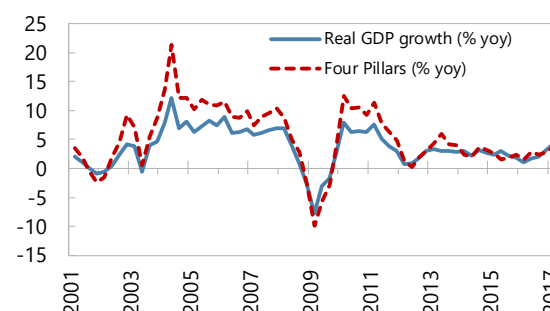


SYNCHRONIZATION OF HONG KONG SAR'S BUSINESS CYCLES WITH THE US AND MAINLAND CHINA¹

While Hong Kong SAR's business cycle remains more synchronized with that of the United States, Mainland China plays an increasingly important role in driving idiosyncratic developments in key sectors. This note sheds light on how external developments affect growth prospects of Hong Kong SAR.

1. Understanding the synchronization of Hong Kong SAR's business cycles with those of major trading partners is important for assessing its growth prospects. Hong Kong SAR is a small highly-open economy, buffeted not only by domestic factors, but also external developments. Although it has traditionally been predominately affected by spillovers from the U.S., in recent years, Hong Kong SAR's economy has become increasingly integrated with that of Mainland China. The impact of spillovers on Hong Kong SAR's business cycle and sectoral developments is important for assessing Hong Kong SAR's growth prospects. We consider developments in the Four Pillar industries which include financial services, tourism, trading and logistics, and professional services and other producer services, accounting for 57 percent of GDP in 2015.² The correlation between headline real GDP growth and Four Pillars industries was 0.97 during 2001Q1 to 2017Q1.

Hong Kong SAR: Real GDP Growth and Estimated Growth of Four Pillar Industries



Sources: C&SD; and IMF staff calculations.

A. Hong Kong SAR's Business Cycle Remains More Synchronized with that of the US

2. The sectoral growth of Four Pillar industries can be decomposed into a common shock and idiosyncratic sectoral shocks (using principal components).³ The first principal component of the growth rates of Four Pillar industries, labeled henceforth as the "common shock", has moved closely with headline GDP. We label the residual of each sector's growth after deducting the common shock as a "sectoral shock", which tracks idiosyncratic developments in the corresponding

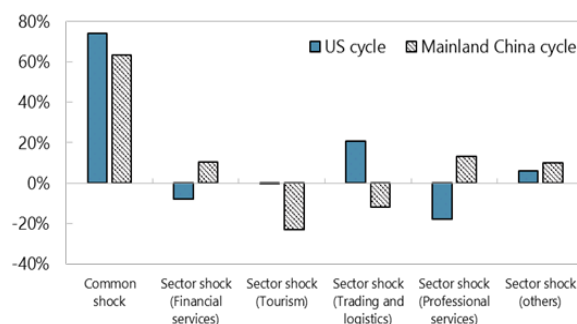
¹ Prepared by Daniel Law.

² The official data of Four Pillars' growth is only available at annual frequency up to 2015. The data series of Four Pillars are extended using weighted averages of chain-linked growth rates of industries in the breakdown of quarterly real GDP growth by industry.

³ In line with HKMA analysis (see for example, Box 3 of HKMA's semiannual Financial Stability Report, September 2016 issue). The Annex provides details on the methodology.

sector. The correlation between the common shock and cyclical swings in the U.S. and Mainland China were 0.74 and 0.63 respectively, during 2001Q1-2017Q1, showing significant unconditional bilateral spillovers from both the U.S. and Mainland China. The magnitudes of the correlation of each sector with either U.S. or Chinese business cycles were generally smaller and their signs varied across sectors. In a multivariate regression to explain the common shock, the coefficients of the cyclical swings in the U.S. and Mainland China were both significant at 0.69 and 0.46 respectively. This preliminary evidence suggested that Hong Kong SAR has a more synchronized cycle with the U.S. than with Mainland China.

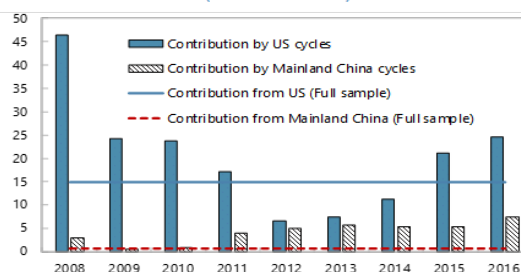
Correlations between shocks in Hong Kong's growth and cycles in US and Mainland China



Sources: C&SD; and IMF staff calculations.

3. Further analysis confirms that Hong Kong SAR's business cycle is still more synchronized with the U.S. In the literature, He, Liao and Wu (2015)⁴ tested Hong Kong's growth synchronization with a structural VAR model based on the permanent-income hypothesis, and found that the transitory component of income in Hong Kong SAR remains more driven by that of the U.S. while the permanent component of income (or trend growth) in Hong Kong SAR is more driven by that of Mainland China. We used the reduced-form VAR model, and the variance decomposition provides direct information on the relative contribution from each external business cycle.⁵ For the full sample of 2001Q1-2017Q1, headline growth (as proxied by the common shock) was more affected by cyclical swings in the U.S., which accounted for 15 percent of the variation. The impact of cyclical swings in Mainland China was much smaller at only 0.6 percent.

Contribution of Variation in HK's Business Cycle from US and Mainland China (Common Shock)



Sources: C&SD; and IMF staff calculations.

4. However, spillovers from Mainland China have been growing. We estimated 32-quarter rolling window versions of the 3-variable VAR to examine the extent of spillovers over time. The effect of the U.S. was the highest at 46 percent during the global financial crisis in 2008, then fell gradually to 6.6 percent in 2012, and rebounded subsequently to 25 percent in 2016. On the other

⁴ See He, Dong, Wei Liao and Tommy Wu, 2015, "Hong Kong's Growth Synchronization with China and the U.S.: A Trend and Cycle Analysis", IMF Working Paper WP/15/82.

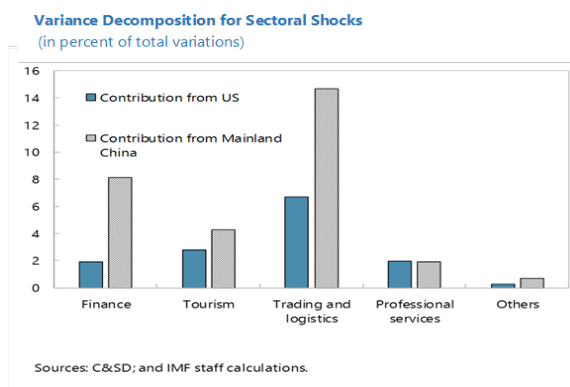
⁵ We used a 3-factor VAR model which included the common shock, cyclical swings in the U.S. and China with 1-quarter lags. The variance decomposition was obtained using the average contribution across two Cholesky decompositions: one where shocks to Mainland China's business cycle are not contemporaneously affected by the U.S. and common shock and another where shocks to the U.S. business cycle are not contemporaneously affected by Mainland China or the common shock.

hand, influence from Mainland China has been growing though it is still smaller; it jumped to 4 percent in 2011 from (below 1 percent) and then rose gradually to 7.4 percent in 2016.

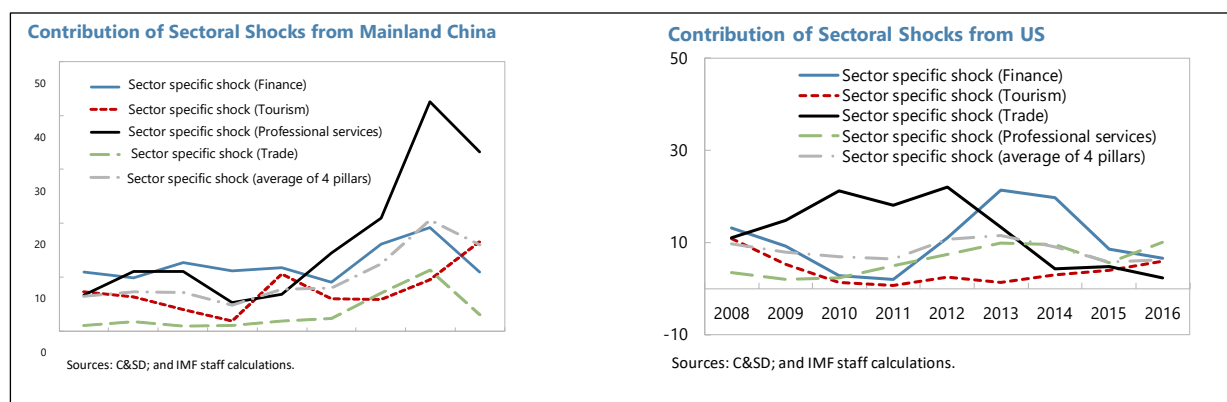
B. Increasing Contribution of Mainland China in Key Sectors

5. Analysis showed that China is playing a bigger role in driving idiosyncratic developments in certain sectors in Hong Kong SAR.

We assess whether Mainland China or the U.S. cycles drive sectoral shocks, by replacing the common shock with sectoral shocks in the 3-factor VAR models. The results suggest that idiosyncratic developments of the key sectors are increasingly driven by Mainland China, reflecting greater integration. Financial services have benefited from gradual capital account liberalization in Mainland China. Tourism stayed weak on stalled momentum in Mainland tourist arrivals and their spending, while trading and logistics was sensitive to external demand from Mainland China.



6. The influence of Mainland China on the idiosyncratic developments in the key sectors rose in recent years, especially on trading and logistics. Using variance decompositions over rolling 32-quarter samples, one can see the decline in the role of the U.S. business cycle and the rising importance of Mainland China. On average, the contribution from cyclical swings in the U.S. has declined from the peak at 11 percent in 2013 to about 6 percent in 2016. On the other hand, the influence from cyclical swings in Mainland China rose from 5 percent in 2011 to the peak at 21 percent in 2015, easing slightly to 16 percent in 2016. The trading and logistics sector witnessed the sharpest increase among the Four Pillars with 33 percent of variation from Mainland China in 2016, compared to 3 to 17 percent for other three Pillars. For tourism, the contribution from both US and Mainland China increased in recently years.



7. In summary, based on analysis of common and sectoral shocks of pillar industries, Hong Kong SAR's business cycles are still more related to the cyclical swings in the U.S. Mainland China's influence has grown substantially, and cyclical swings in Mainland China

contributed more to idiosyncratic sectoral developments in Hong Kong SAR, in particular in trading and logistics and tourism. These results are in line with He et al (2015) that transitory shocks from the U.S. remain a major driving force behind Hong Kong SAR's business cycle fluctuations, while permanent shocks from Mainland China have a larger impact on Hong Kong SAR's trend growth.

Annex I. Technical Details on the Synchronization of Hong Kong SAR's Business Cycles with the US and China

Data on Four Pillars and Headline GDP

The data series of Four Pillars were only available annually up to 2015 and were extended to span from 2000Q1 to 2017Q1 using weighted averages of chain-linked growth rates of industries in the breakdown of quarterly real GDP growth by industry (with different classification from Four Pillars).

Decomposition of Common Shock and Sectoral Shock

Following HKMA half-yearly financial stability report (September 2016 issue), we decomposed the growth of Four pillars into common shock and sectoral shock,

Sectoral growth = constant + β (common shock) + sectoral shock, and

Common shock = first principal component for Four Pillar industries and the residual "Others" industries

Correlation Analysis

To conduct correlation analysis, we need to estimate the cyclical swings in the US and China respectively. Therefore, we use HP filter on the quarterly real GDP growth of the US and China respectively to get the residuals. The residuals are assumed to be the cyclical swings in the US and China.

For the correlation analysis, we calculated the correlation coefficients of the cyclical swings in the US with common shock and sectoral shocks respectively and then the cyclical swings in the US is replaced by the cyclical swings in China. The sample period is 2001Q1 – 2017Q1.

Multivariate Regression

We regressed the common shock on the cyclical swings in the U.S. and China to get a sense on their sensitivity to the business cycle:

Common shock = $\alpha + \beta_1$ (cyclical swings in the U.S.) + β_2 (cyclical swings in China) + ε

VAR Model for Common Shock

VAR model is a useful tool to identify the source and response, especially from the distribution of variance decomposition. In order to understand the source of Hong Kong's business cycle (common shock as proxy). For sample period of 2001Q1 – 2017Q1, we used a simple 3-factor VAR model (with 1-quarter lag based on SIC):

$$X_t = A + B X_t + \varepsilon_t, \text{ and}$$

$X_t = (\text{common shock, US cycle, China cycle})$, A is a vector of constant, B is a vector of slope coefficients and ε_t is a vector of error terms.

The variance decomposition is obtained from Cholesky decomposition, which is affected by the ordering of variables. Therefore, we also ran the model in the variable order of (common shock, China cycle, US cycle) and take the averages for the two in the variance decomposition. A 10-quarter forecast horizon was used.

As the common shock is simply first principal component (the contribution from common shock is a scalar factor multiplied by the first principal component), we did not run the model for each of the Four Pillars as they should give the same variance decomposition.

In order to know how the contribution evolved over time, we ran the models using rolling 32-quarter samples ending 2008Q4, 2009Q4, 2010Q4, 2011Q4, 2012Q4, 2013Q4, 2014Q4, 2015Q4 and 2016Q4 respectively.

VAR Models for Sectoral Shocks

Similar to the VAR for common shock, we replaced the common shock by sectoral shocks to examine the impact of cyclical swings in the US and China on Hong Kong's idiosyncratic sectoral developments.

$$X_t = A + B X_t + \varepsilon_t, \text{ and}$$

$X_t = (\text{sectoral shock of industry } i, \text{ US cycle, China cycle})'$, A is a vector of constant, B is a vector of slope coefficients and ε_t is a vector of error terms.

We also ran the models for each of the Four Pillar industries with variable order of (sectoral shock of industry i , China cycle and US cycle) and with rolling 32-quarter samples. The variance decomposition is from Cholesky decomposition and a 10-quarter forecast horizon was used.

References

He, D., W. Liao and T. Wu, 2015, “Hong Kong's Growth Synchronization with China and the U.S.: A Trend and Cycle Analysis,” IMF Working Paper WP/15/82.

Hong Kong Monetary Authority, 2016, “Box 3. The unemployment rate of Hong Kong: The effects of aggregate and sectoral channels,” Half-Yearly Monetary and Financial Stability Report (September 2016 issue).